

MICROECONOMICS

Assignment for Tutorial 1 - September 22, 2025

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Instructions

- You should review and complete these exercises before each session.
- You may only attend the tutorial session to which you have been assigned. Group 1 is on Monday 16h30 - 18h30 (room 115), Group 2 is on Monday 18h30 - 20h30 (room 115).
- If you are not familiar with how to use the Lagrangian or Kuhn-Tucker methods, please consult the slides from the Maths Refresher and the following textbooks:
 - Simon, C., and Blume, L. 1994. *Mathematics for economists*. Norton.
 - Chiang, A., and Wainwright, K. 2005. *Fundamental Methods of Mathematical Economics*. McGraw-Hill Education
 - Dowling, E.T., 2011. *Schaum's Outline of Introduction to Mathematical Economics*. McGraw-Hill Education.

1 Exercises on unconstrained optimisation

Reminder: The first order condition of x^* to be a maximum (minimum) of a function f of one variable is $f'(x^*) = 0$. x^* is then a critical point of f . For a function of n variables, the partial derivatives of f are set to zero.

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

$\nabla f(x)$ is the gradient, a column vector, where the i -th element is the partial derivative of $f(x)$ with respect to x_i . x^* is a critical point of f if and only if $\nabla f(x^*) = 0$.

1. (FOCs) Find the critical points of:

- $f(x) = 2x_1^2 - x_2^2 - 12x_1 - 12x_2$
- $f(x) = x_1^2 - 8x_1x_2 + x_2^2 - 4x_1 + 4x_2$
- $f(x) = 2x_1x_2 - 4x_1 + 2x_2 - 3$

2 Exercises on constrained optimisation

Reminder 1: in economics, optimization usually implies the existence of certain constraints, be it for the distribution of goods, in production or consumption. The mathematical formulation of such problem is as follows:

$$\max f(x_1, x_2, \dots, x_n) \text{ with } (x_1, x_2, \dots, x_n) \in R^n$$

subject to

$$\begin{aligned} g_1(x_1, x_2, \dots, x_n) &= b_1, \dots, g_k(x_1, x_2, \dots, x_n) = b_k \\ h_1(x_1, x_2, \dots, x_n) &\leq c_1, \dots, h_k(x_1, x_2, \dots, x_n) \leq c_k \end{aligned}$$

where f is called the objective function and $g_1, \dots, g_k$ and $h_1, \dots, h_k$ the constraint functions. The g_j are equality constraints, the h_j are inequality constraints.

Equality constraints are easy to deal with, as maximum or minimum of the objective function must also lie on the constraint function. Converting the constrained optimization problem on n variables and k constraints into an unconstrained optimization problem of $n + k$ variables, one can define the Lagrangian function as:

$$L(x_1, x_2, \dots, x_n, \lambda_1, \lambda_2, \dots, \lambda_k) = f(x_1, x_2, \dots, x_n) - \sum_k \lambda_i (g_i(x_1, x_2, \dots, x_n) - b_i)$$

λ_i are called the Lagrange multipliers. The optimization for x^* is then very simple: take the partial derivatives and setting them equal to zero.

$$\begin{aligned} \frac{\partial \mathcal{L}}{\partial x_1}(\mathbf{x}^*, \lambda^*) &= \frac{\partial f}{\partial x_1} - \sum_k \lambda_i \frac{\partial b_i}{\partial x_1} = 0 \\ &\vdots \\ \frac{\partial \mathcal{L}}{\partial x_n}(\mathbf{x}^*, \lambda^*) &= \frac{\partial f}{\partial x_n} - \sum_k \lambda_i \frac{\partial b_i}{\partial x_n} = 0 \\ \frac{\partial \mathcal{L}}{\partial \lambda_1}(\mathbf{x}^*, \lambda^*) &= g_1(x_1, x_2, \dots, x_n) - b_1 = 0 \\ &\vdots \\ \frac{\partial \mathcal{L}}{\partial \lambda_k}(\mathbf{x}^*, \lambda^*) &= g_k(x_1, x_2, \dots, x_n) - b_k = 0 \end{aligned}$$

Inequality constraints are more challenging, because it is not clear which constraints will be binding and which will not before the optimization. Inequality constraints define the boundary of a region over which we seek to optimize the function. The optimum could therefore be in the interior of the region, or somewhere along one of the constraints. As with an equality constraint, the constrained optimization constraint is transformed into an unconstrained optimization constraint for easier handling:

$$L(x_1, x_2, \dots, x_n, \lambda_1, \dots, \lambda_m, s_1, \dots, s_m) = f(x_1, x_2, \dots, x_n) - \sum_m \lambda_i (g_i(x_1, x_2, \dots, x_n) + s_i^2 - b_i)$$

where for each inequality constraint $g_i(x_1, \dots, x_n) \leq b_i$ we define a slack variable s_i^2 , for which the expression $g_i(x_1, \dots, x_n) \leq b_i - s_i^2$ would hold with equality. These slack variables s_i^2 capture how close the constraint comes to binding. The variable is squared to ensure that the slack is positive. In order to find the critical points, we now need to take the partial derivatives with respect to each x , λ and s . This results in $n + 2m$ equations with $n + 2m$ unknowns:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x_1}(\mathbf{x}^*, \lambda^*) &= \frac{\partial f}{\partial x_1} - \sum_m \lambda_i \frac{\partial b_i}{\partial x_1} = 0 \\
&\vdots \\
\frac{\partial \mathcal{L}}{\partial x_n}(\mathbf{x}^*, \lambda^*) &= \frac{\partial f}{\partial x_n} - \sum_m \lambda_i \frac{\partial b_i}{\partial x_n} = 0 \\
\frac{\partial \mathcal{L}}{\partial \lambda_1}(\mathbf{x}^*, \lambda^*) &= g_1(x_1, \dots, x_n) + s_1^2 - b_1 = 0 \\
&\vdots \\
\frac{\partial \mathcal{L}}{\partial \lambda_m}(\mathbf{x}^*, \lambda^*) &= g_m(x_1, \dots, x_n) + s_m^2 - b_m = 0 \\
\frac{\partial \mathcal{L}}{\partial s_1}(\mathbf{x}^*, \lambda^*) &= 2s_1\lambda_1 = 0 \\
&\vdots \\
\frac{\partial \mathcal{L}}{\partial s_m}(\mathbf{x}^*, \lambda^*) &= 2s_m\lambda_m = 0
\end{aligned}$$

Reminder 2: The Kuhn-Tucker conditions provide a method that often makes it easier to calculate the critical points and identify points that are (local) maxima than the vanilla approach described above. The setup is similar, but treats the Lagrangian as if it were an optimization with equality constraints:

$$L(x_1, x_2, \dots, x_n, \lambda_1, \dots, \lambda_m) = f(x_1, x_2, \dots, x_n) - \sum_m \lambda_i (g_i(x_1, x_2, \dots, x_n) - b_i)$$

The Kuhn-Tucker conditions are then:

$$\begin{aligned}
\frac{\partial \mathcal{L}}{\partial x_1} &\leq 0, \dots, \frac{\partial \mathcal{L}}{\partial x_n} \leq 0 \\
\frac{\partial \mathcal{L}}{\partial \lambda_1} &\geq 0, \dots, \frac{\partial \mathcal{L}}{\partial \lambda_m} \geq 0 \\
x_1 &\geq 0, \dots, x_n \geq 0 \\
\lambda_1 &\geq 0, \dots, \lambda_n \geq 0 \\
x_1 \frac{\partial \mathcal{L}}{\partial x_1} &= 0, \dots, x_n \frac{\partial \mathcal{L}}{\partial x_n} = 0 \\
\lambda_1 \frac{\partial \mathcal{L}}{\partial \lambda_1} &= 0, \dots, \lambda_m \frac{\partial \mathcal{L}}{\partial \lambda_m} = 0
\end{aligned}$$

The last two sets of conditions are analogous to the complementary slackness conditions. To minimize the function $f(x_1, \dots, x_n)$, maximize the function $-f(x_1, \dots, x_n)$.

1. Solve all the following exercises.

- Minimize $(x_1 - 1)^2 + (x_2 - 2)^2$ subject to $x_1 \leq x_2$; and $x_1 + 2x_2 \leq 2$.
- Minimize $12x_1 - 5x_2 + 3x_3$ subject to $x_1 + 2x_2 - x_3 = 5$; $x_1 - x_2 \geq -2$; and $2x_1 - 4x_2 \leq 12$.
- Minimize $-2(x_1 - 2)^2 - x_2^2$ subject to $x_1^2 + x_2^2 \leq 25$, and $x_1 \geq 0$.
- Find the max and min of $f(x, y, z) = x + y + z^2$ subject to $x^2 + y^2 + z^2 = 1$; and $y = 0$.
- Maximise $f(x, y, z) = yz - xz$ subject to $y^2 + z^2 = 1$; and $xz = 3$.
- Minimize $f(x_1, x_2) = (x_1 - 4)^2 + (x_2 - 4)^2$ subject to $2x_1 + 3x_2 \geq 6$; $-3x_1 - 2x_2 \geq -12$; and $x_1, x_2 \geq 0$.

Problem 1 - First glimpse: UMP

- (a) Consider a utility maximization problem with two goods (x and y with prices $p_x = 2$ and $p_y = 1$). Write down the budget constraint for an income equal to 10.
- (b) Plot the budget constraint.
- (c) Consider the following utility function: $u(x, y) = \sqrt{x} + \sqrt{y}$. Plot three indifference curves for this function. [**Hint:** the indifference curve at level k is defined as : $u(x, y) = \sqrt{x} + \sqrt{y} = k$.]
- (d) Without any algebra, draw the solution to the maximization problem.
- (e) Now consider a different utility function: $u(x, y) = x + y$. Plot three indifference curves for this function.
- (f) Without any algebra, draw the solution to the maximization problem.

Problem 2 - UMP again

Consider a standard maximization problem with the following utility function:

$$u(x, y) = \sqrt{x} + \sqrt{y}$$

The price of good x is p_x , the price of good y is p_y , and income is denoted with M . Assume that $x > 0$ and $y > 0$ unless otherwise stated.

- (a) Compute $\frac{\partial u(x, y)}{\partial x}$ and $\frac{\partial^2 u(x, y)}{\partial x^2}$. Is the utility function increasing in x ? Is the utility function concave in x ?
- (b) The consumer maximize utility subject to a budget constraint. Write down the maximization problem of the consumer with respect to x and y . Explain briefly why the budget constraint is satisfied with equality.
- (c) Write down the Lagrangean function.
- (d) Write down the FOCs for this problem.
- (e) Derive the Marshallian demand function of both x and y .
- (f) Do the Marshallian demands satisfy the positive constraints, $x > 0$ and $y > 0$?